OpenStack Network Architecture

Overview

The Virtual SOC environment is built on a segmented OpenStack network topology designed to replicate enterprise-grade routing, segmentation, and security zones. Each network and router plays a specific role that enables traffic control, monitoring, and secure access.

Architectural Goals

- Provide isolated network zones for SOC components.
- Ensure traffic must pass through pfSense to reach internal networks.
- Support SIEM, NIDS, attacker, and victim communications.
- Provide remote-access pathways without exposing internal subnets.
- Maintain compatibility with OpenStack Neutron routing.

Network Topology Summary

```
ext_net ── ex-router ── transit-net (/29) ── pfSense WAN
                                     │
                                     └── internal-net (/24) ── victim hosts /
servers
```

Networks

1. External Network (ext_net)

- Provided by OpenStack.
- Supports floating IPs.
- Used as:
- Upstream gateway for pfSense (via ex-router).
- Remote access entry point.
- WireGuard tunnel endpoint.

2. Transit Network (transit-net)

- Subnet: **10.250.2.0/29**
- Created using CLI:
- `openstack network create transit-net`
- `openstack subnet create --network transit-net --subnet-range 10.250.2.0/29 --gateway 10.250.2.1 transit-subnet`

Purpose

- Provides a buffer segment between OpenStack routing and pfSense WAN.
- Allows traffic shaping and NAT control.
- Enables OpenStack to understand downstream routing.

3. Internal Network (LAN / internal-net)

- Subnet: **105.45.5.0/24**
- Default gateway: **105.45.5.1** (pfSense LAN IP)
- Hosts:
- Victim machines
- Metasploitable 3
- RDP client (temporary)
- Wazuh agents

DHCP

- Provided by pfSense: 105.45.5.4–105.45.5.254

Routers

ex-router

- Connects **ext_net** ↔ **transit-net**
- Commands used:
- `openstack router create ex-router`
- `openstack router add subnet ex-router transit-subnet`
- `openstack router set --external-gateway ext_net ex-router`

Static Route Configuration

To route traffic to the isolated internal network:

```
openstack router add route --route destination=105.45.5.0/24,gateway=10.250.2.5
ex-router
```

Where **10.250.2.5** is pfSense WAN.

pfSense Interface Mapping

OpenStack binds IPs to ports before pfSense boot. - Obtain port assignments:

```
openstack port list --server <pfsense>
```

- Rename for clarity:

```
openstack port set --name pfw <wan-port-id>
openstack port set --name pfl <lan-port-id>
```

- Reassign LAN IP for gateway role:

```
openstack port set --no-fixed-ip pfl
openstack port set --fixed-ip subnet=lantest-subnet,ip-address=105.45.5.1 pfl
```

Connectivity Validation

Upstream (LAN → Transit → ext_net)

- Ping 8.8.8.8 (DNS)
- Ping/Traceroute to floating-IP hosts

Downstream (ext_net → pfSense → internal-net)

- Requires static routing
- Verified after ex-router route insertion

Lateral Traffic

- Confirmed between rdp-client and internal hosts
- Ensured pfSense sees all traffic from attacker → victim

RDP Access Method

Because pfSense initially had no bastion: 1. rdp-client temporarily bridged into LAN 2. RDP / XFCE installed 3. Used to access pfSense GUI at https://105.45.5.1

This was retired once WireGuard was deployed.

Attacker Routing (Kali)

Persistent static routes added:

```
post-up ip route replace 10.250.2.0/29 via <ex-router-ip>
post-up ip route replace 105.45.5.0/24 via <ex-router-ip>
```

Ensures all RFC1918 traffic flows through pfSense.

SSH Bastion Path

```
ext_host → pfSense (ProxyJump) → internal hosts
```

Enabled by pfSense SSH with agent forwarding.

Summary

The OpenStack network layer forms the backbone of the Virtual SOC environment. By isolating networks into ext, transit, and LAN segments—and enforcing all downstream access through pfSense—the topology ensures that: - All traffic is observable. - NIDS sensors have full visibility. - SIEM receives logs from every path in the chain. - Remote operators can securely interact with the environment without exposing critical subnets.